

Inference of Protein Secondary Structure with HMMs and BayesFlow

Simulation-Based Inference

Armin Maddah Asl Samaneh Naghdbishi Kiana Ebrahimi Farbod Geri Lila

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TU Dortmund University

Course Instructor: Prof. Dr. Paul Bürkner

Introduction & Modeling Idea

The Problem

Predict per-position secondary structure
from variable-length amino acid sequences.

Two-State HMM

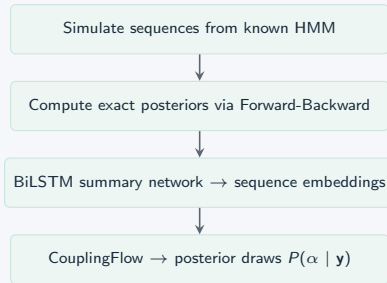
State 0 = α -helix

State 1 = other (β -sheets & coils)

Modeling Goal

Approximate $P(S_t = \alpha \mid \mathbf{y})$ at every position t
using **BayesFlow** neural posterior estimation.

Overall Pipeline



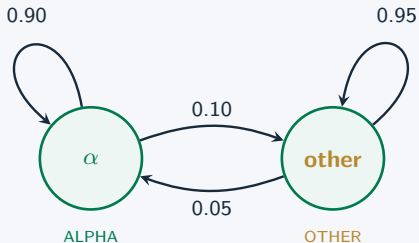
Validation

Compare to experimental ground truth:

Human insulin **1A7F** (chain A: 21 aa, chain B: 29 aa).

The 2-State Hidden Markov Model

State Transitions



Deterministic start: $P(S_1 = \text{other}) = 1.0$, $P(S_1 = \alpha) = 0.0$

Highly persistent: α stays α (90%), other stays other (95%)

Selected Emission Contrasts

Amino acid	α -helix	other	Role
A (Alanine)	12%	6%	Strong former
E (Glutamic)	9%	5%	Former
P (Proline)	2%	6%	Breaker
G (Glycine)	4%	9%	Breaker

Full 20×2 table in written report.

Prior Dynamics

Stationary distribution:

$$\pi(\alpha) = 0.05 / (0.05 + 0.10) = 1/3 \approx 33\%$$

$$\pi(\text{other}) \approx 67\%$$

Prior strongly favors **non-helical** states ($\sim 67\%$).

Data Generation & Analytical Targets

Simulator

Sequential Markov sampling of hidden states.

Independent randomized emission draws.

Random sequence lengths: 10–60 aa.

100% simulated — no real proteins.

Target Computation

Exact posteriors via Forward-Backward.

`hmmlearn.CategoricalHMM` — manually

configured, never `.fit()`.

Verification

Targets verified against independent

pure-NumPy FB solver. Confirms:

$P(S_0 = \alpha) \equiv 0$ (position 0 = seq. position 1).

Real-world validation: human insulin **1A7F** prepared here — sequence + experimental α -labels (used in slide 11).

On-demand Generation

`generate_dataset(n, ...)` produces
any number of (sequence, posterior,
states) tuples. Downstream code

chooses the split sizes.

Variable-Length by Design

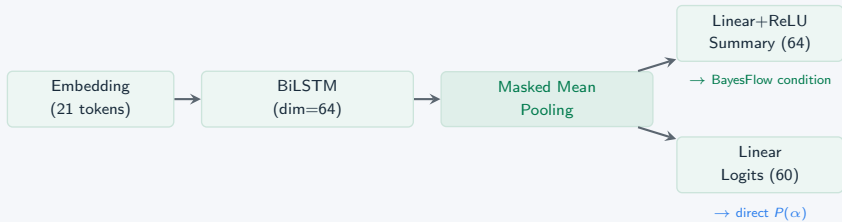
Sequences kept **ragged** (10–60 aa),
each stored with its own length.
The data layer does **not pad** — padding
& masking are handled downstream by
the summary network.

Storage

`.npz` format: concatenated arrays +

lengths vector. No object arrays.

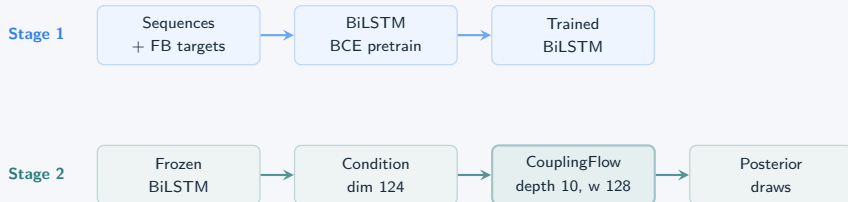
The Summary Network — BiLSTM



1. BiLSTM captures **sequential** dependencies — permutation-invariant nets fail here.
2. Padding token 20 & masked loss → handles variable-length sequences.
3. Two-stage: BiLSTM pretrained on BCE → frozen → CouplingFlow on top.

Two-Stage Training & CouplingFlow

Training Pipeline

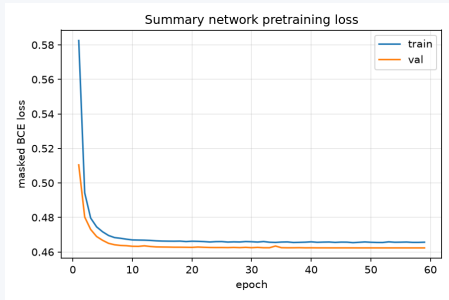


Key Design Decisions

1. BiLSTM pretrained on masked BCE, then **frozen** → condition (dim 124 = 64 pooled + 60 logits).
2. CouplingFlow: depth 10, hidden 128, affine, actnorm → posterior draws.
3. **Model selection:** **not** flow val_loss → rank by masked MAE & correlation (selected depth 10).

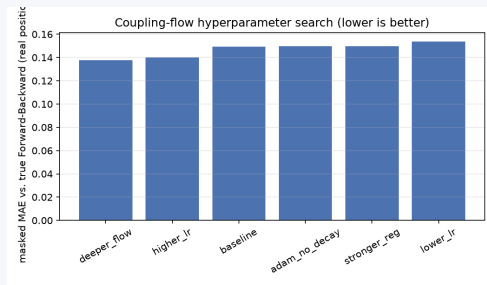
Training Diagnostics & Model Selection

Stage-1 pretraining loss



AdamW, grad-clip 1.0, early stopping (patience 8).

CouplingFlow hyperparameter search



6 configs, ranked by masked MAE — not flow loss → depth 10.

Converged, not stuck. Soft targets $\Rightarrow$ BCE has an **entropy floor**; it cannot reach zero.

Validation: floor $\bar{H} = 0.4621$ achieved = **0.4623** $\Rightarrow$ **0.0002 from the theoretical limit.**

Diagnostics: Adapted Simulation-Based Calibration

Why Standard SBC Does Not Apply

Standard SBC (Talts et al., 2018):

$\theta \sim p(\theta)$ (draw parameter)

$x \sim p(x | \theta)$ (simulate data)

Rank θ_{true} in posterior

→ Ranks must be **uniform**.

The Problem

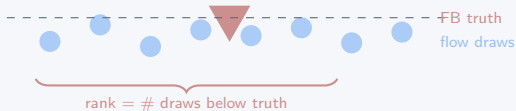
HMM parameters are **fixed**.

$P(\alpha | \mathbf{y})$ is a **deterministic** function of the sequence.

No θ to sample from a prior.

→ Standard SBC is impossible.

Our Adaptation: Position-wise SBC



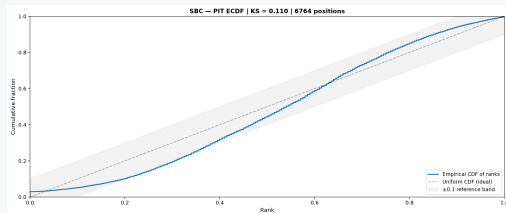
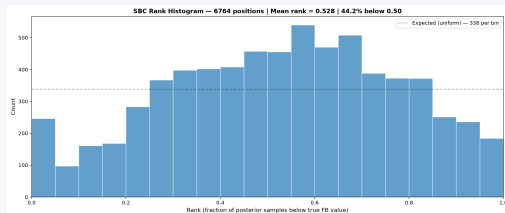
Key idea: Instead of ranking an unknown parameter, rank the **known** FB target within the flow's posterior draws at **each sequence position**.

6,764 positions tested → rank histogram, PIT ECDF, KS statistic.

Tests: Does the flow's posterior **envelope** the true value uniformly?

≡ Is the posterior **calibrated per quantile**?

Calibration Results — Adapted SBC Diagnostics



Rank histogram (left)

PIT ECDF with KS statistic (right)

6,764 positions

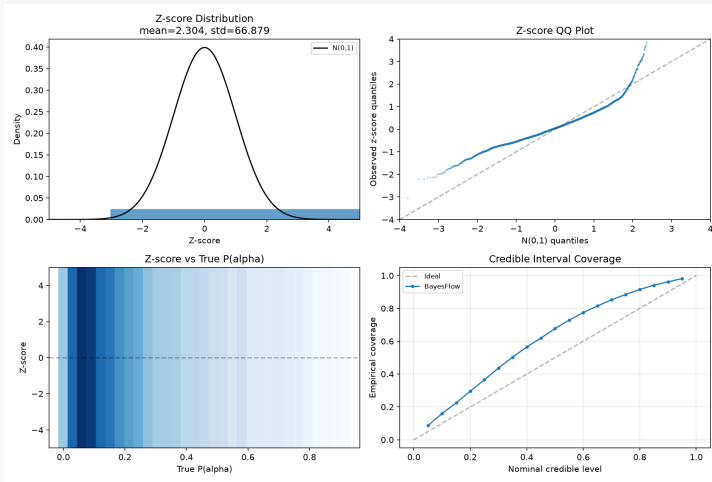
Adapted SBC Results

- Mean Rank = **0.53** (Near-uniform — good calibration)
- **44%** of ranks below 0.50
- KS Distance = **0.110** (Moderate deviation)

The Spatial Cause

Position 0 (deterministic start) and padded boundaries are **exact zeros**. Normalizing flows are continuous — they struggle to represent **point masses**.

Calibration Results — Z-Score & Coverage Diagnostics



Key Metrics

Raw Z-std = **36.2**

(pos-0 degeneracy)

Trimmed (1.1%):

mean = **-0.19**

std = **0.94**

Coverage:

94.2% vs. 90%

Well-Calibrated

Mean near zero

(-0.19)

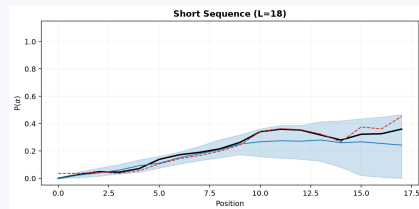
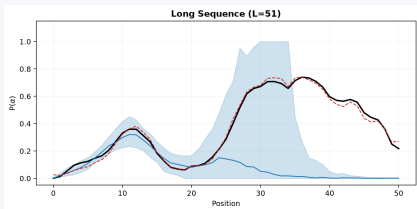
+ std < 1.0 (0.94)

→ posteriors

well-spread

+ coverage 94.2%

Synthetic Evaluation & Posterior Contraction



Black = True $P(\alpha)$ Blue band = BayesFlow 90% CI Red dashes = BiLSTM point

Accuracy Metrics

Model	MAE	r
Baseline (sequence-blind)	0.162	0.200
BayesFlow (median)	0.057	0.778
BiLSTM (point)	0.004	0.9997

6,764 positions. Both beat the naive baseline.

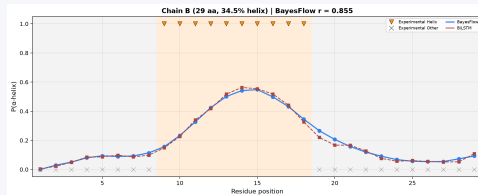
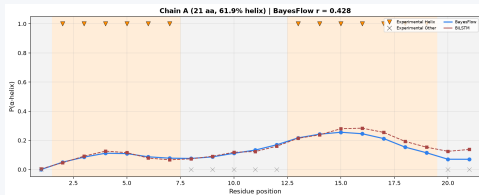
Why BiLSTM Wins

$P(\alpha \mid \mathbf{y})$ is a **deterministic** function of the sequence. A direct network learns it with near-zero error. A flow models a residual density.

Why We Still Use BayesFlow

BiLSTM yields **no uncertainty**. BayesFlow provides full amortized posteriors $\rightarrow$ credible intervals, SBC, z-scores, and coverage diagnostics.

Real-World Validation — Human Insulin (1A7F)



Orange = Experimental Helix Grey = Experimental Other Blue = BayesFlow Red = BiLSTM

The Statistical Mechanism

Chain B — Agreement ($r=0.85$)

34.5% true helix $\approx \pi=33\%$ prior. Prior and evidence agree $\rightarrow$ high accuracy.

Chain A — Conflict ($r=0.43$)

61.9% true helix $\gg 33\%$ prior. Severe prior-data conflict over only 21 residues.

Both beat sequence-blind baseline ($MAE=0.162$, $r=0.200$).

Key Proof — Model Limitation

Exact FB also fails on Chain A

Running the exact analytical Forward-Backward solver on Chain A achieves only $r \approx 0.40$.

$\rightarrow$ The failure is a **model limitation** of the 2-state HMM, **not** a neural network issue.

No amount of training can exceed the HMM's ceiling.

Take-Home Messages & Future Outlook

1. **Sequence Matters** — BiLSTM networks capture sequential dependencies where permutation-invariant architectures fail.
2. **Beyond Point Estimates** — BayesFlow amortizes full posterior densities, enabling SBC, z-scores, and coverage diagnostics.
3. **Prior Typicality** — Real-world validation reveals the necessity of checking prior-data conflicts (e.g., Insulin Chain A).

Limitations & Potential Fixes

The HMM: Expand to 3 states (α , β , coil) and infer transition parameters under priors.

The Flow: Pass targets through a logit transform before training $\rightarrow$ bound draws in $(0, 1)$.

Group Members & Code

Armin Maddah Asl

Samaneh Naghdbishi

Kiana Ebrahimi Farbod

Geri Lila

[github.com/armin2080/
protein-secondary-structure-npe](https://github.com/armin2080/protein-secondary-structure-npe)